

VIENTIANE WATTAY INTL, Lao People's Democratic Republic

WMO: 489400

Lat: 17.988N

Lon: 102.563E

Elev: 172

StdP: 99.28

Time Zone: 7.00 (E07)

Period: 03-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	14.3	16.0	8.5	7.0	19.6	10.4	8.0	19.3	5.2	22.8	4.3	22.4	1.3	0	0.291

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
4	10.3	37.4	26.2	36.2	26.0	35.1	25.8	28.1	33.7	27.6	32.8	27.2	32.1	1.6	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
26.8	23.0	30.2	26.2	22.1	29.5	26.1	21.9	29.3	91.4	33.8	89.4	33.1	87.6	32.4	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
4.6	3.7	3.2	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	27.3	23.1	25.6	27.9	29.8	29.4	29.1	28.5	28.2	28.3	27.9	26.1	23.7
	DBStd	2.87	2.55	2.53	2.61	2.15	1.71	1.42	1.45	1.33	1.10	1.30	2.14	2.37
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	4	3	0	1	0	0	0	0	0	0	0	0	1
	CDD10.0	6312	405	436	554	594	601	572	572	563	550	556	483	426
	CDD18.3	3273	149	202	297	344	342	322	314	305	300	297	233	168
	CDH23.3	35019	1088	1921	3356	4391	4052	3769	3421	3226	3258	3157	2137	1243
CDH26.7	14751	331	852	1658	2339	1902	1608	1298	1148	1213	1225	801	376	
Wind	WSAvg	1.5	1.4	1.5	1.5	1.6	1.5	1.5	1.5	1.4	1.4	1.4	1.4	1.5
Precipitation	PrecAvg	1604	10	17	43	95	223	252	278	335	283	100	15	6
	PrecMax	2071	80	120	135	291	520	474	635	637	609	312	107	40
	PrecMin	1311	0	0	1	1	50	67	133	160	122	0	0	0
	PrecStd	220	17	24	35	70	103	97	108	111	111	65	28	10
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	32.9	35.9	37.9	39.8	38.1	36.1	36.0	34.2	34.2	34.1	34.0	32.7
		MCWB	22.0	22.2	24.8	27.2	27.3	27.1	27.0	26.9	27.0	26.1	25.6	24.1
	2%	DB	31.2	34.3	36.3	38.1	36.5	34.9	34.1	33.2	33.2	33.2	32.9	31.2
		MCWB	22.3	23.1	24.8	26.4	27.2	26.9	26.7	26.6	26.7	25.6	24.9	23.3
	5%	DB	30.0	33.1	35.1	36.8	35.1	33.8	33.0	32.4	32.8	32.4	32.0	30.0
		MCWB	22.0	23.1	24.8	26.3	26.9	26.8	26.4	26.4	26.5	25.2	24.3	22.6
	10%	DB	28.8	31.8	33.9	35.1	33.9	32.8	31.9	31.7	31.9	31.9	31.0	28.9
		MCWB	21.3	22.8	24.7	26.0	26.7	26.5	26.3	26.3	26.2	25.0	23.7	21.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	24.6	25.5	27.3	28.6	28.8	28.3	28.0	27.8	28.0	27.3	26.5	25.0
		MCDB	29.6	32.5	34.8	37.3	35.6	33.0	32.4	31.4	31.9	31.5	31.9	30.7
	2%	WB	23.6	24.7	26.5	27.6	28.0	27.7	27.3	27.2	27.4	26.7	25.8	24.1
		MCDB	28.6	31.6	33.9	35.9	34.3	32.2	31.5	31.1	31.3	30.8	30.6	29.3
	5%	WB	22.7	24.1	25.7	27.0	27.4	27.2	26.9	26.9	27.0	26.2	25.3	23.5
		MCDB	27.9	30.8	32.8	34.5	33.3	31.6	31.0	30.7	30.8	30.1	29.7	28.4
	10%	WB	22.1	23.5	25.1	26.4	26.9	26.9	26.6	26.5	26.6	25.8	24.7	22.7
		MCDB	26.8	29.8	31.9	33.2	32.3	31.1	30.3	30.0	30.2	29.5	28.9	27.4
Mean Daily Temperature Range	5% DB	MDBR	10.7	11.5	10.9	10.3	8.7	7.2	6.6	6.5	7.1	7.9	9.5	10.3
		MCDBR	12.0	13.3	12.8	12.2	10.6	8.7	8.5	8.0	8.4	8.9	10.1	11.1
		MCWBR	4.9	4.6	4.4	4.2	3.9	3.0	2.9	3.0	3.1	3.0	3.8	4.4
	5% WB	MCDBR	10.7	11.7	11.6	11.3	9.6	7.7	7.3	7.1	7.5	8.2	9.1	10.4
		MCWBR	4.7	4.7	4.7	4.2	3.9	3.1	2.8	3.0	3.1	3.0	3.8	4.5
Clear-Sky Solar Irradiance	taub	0.465	0.575	0.701	0.666	0.504	0.456	0.445	0.464	0.507	0.545	0.454	0.459	
	taud	2.118	1.841	1.602	1.717	2.169	2.331	2.358	2.286	2.151	2.001	2.232	2.154	
	Ebn at Noon	812	739	665	694	806	837	848	838	803	755	819	804	
	Edh at Noon	149	207	271	242	152	128	125	135	154	174	132	140	
All-Sky Solar Radiation	RadAvg	4.60	5.05	5.14	5.61	5.25	4.81	4.53	4.47	4.62	4.81	4.80	4.45	
	RadStd	0.32	0.28	0.30	0.29	0.39	0.38	0.31	0.32	0.25	0.30	0.20	0.20	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)			
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A			
Regional (10 neighbors)	+0.27	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A			

Nomenclature: See separate page